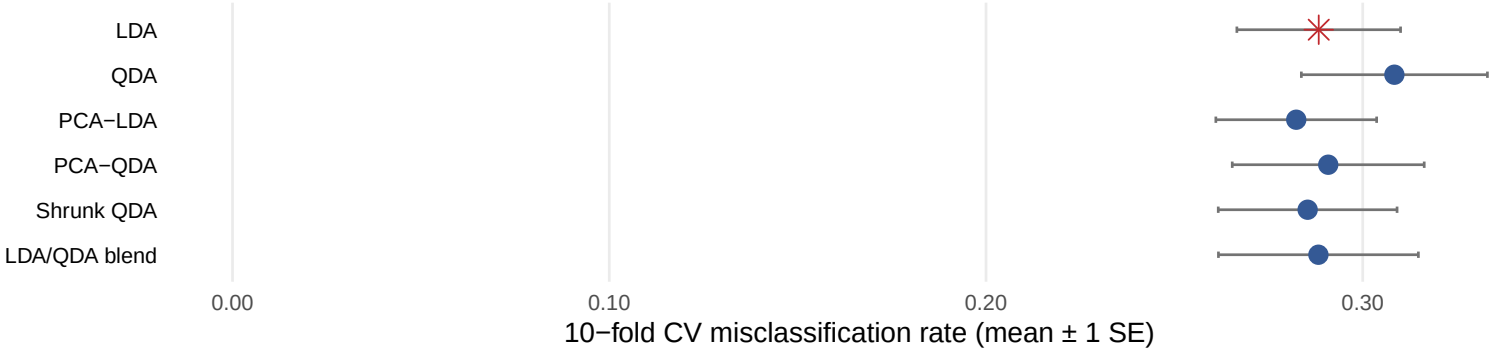
SAheart: LDA/QDA Prediction

Setup and model selection

We analyzed `chd` (0/1) using all nine numeric predictors, with `famhist` coded 1 = Absent and 2 = Present. The data contain 462 people (class counts 0:302, 1:160); the stratified 75/25 split left 347 training and 115 test cases (0:227, 1:120 train; 0:75, 1:40 test). Seed: **2026**.

All candidates used the same stratified 10-fold training splits, empirical class priors, and fold-only preprocessing. Tuned grids were PCA + LDA/QDA (`k=1,...,8`, standardized predictors), QDA covariance shrinkage toward pooled covariance (`lambda=0.1,...,0.9`), and probability blend `(1-w)LDA + wQDA` (`w=0.1,...,0.9`). Ordinary LDA and QDA were untuned baselines.

The lowest-CV-error candidate was **PCA-LDA (k=7)** (0.282); the one-standard-error rule selected the simpler **LDA (ordinary)** (0.288, SE 0.022). Each point below summarizes the best setting in its model family; the selected classifier is marked with an asterisk.



Candidate	Tuned setting	CV error	SE
* LDA	ordinary (selected)	0.288	0.022
QDA	ordinary	0.308	0.025
PCA-LDA	k=7	0.282	0.021
PCA-QDA	k=4	0.291	0.025
Shrunk QDA	lambda=0.5	0.285	0.024
LDA/QDA blend	w=0.5	0.288	0.027

Reserved test-set result

The final classifier was refit on all training observations and evaluated once on the reserved test set. Its training class priors were 0:0.654, 1:0.346. Test confusion matrix (observed rows, predicted columns):

Observed \ Predicted	0	1
0	64	11
1	18	22

Test error was **0.252** (exact 95% binomial interval 0.176–0.342; 29/115 errors).

Findings and limitations

Compared with ordinary LDA (0.288 CV error, SE 0.022) and QDA (0.308, SE 0.025), the selected model's CV error was 0.000 and 0.020 lower, respectively. PCA-LDA's minimum was only 0.006 below LDA, so the one-SE rule favored the simpler baseline; the small differences are not compelling relative to fold variability. Gaussian class distributions and LDA's common-covariance assumption may be wrong; covariance estimates are noisy at this sample size. The test estimate comes from one stratified split. `famhist` is treated as a numeric 1/2 predictor as requested.